

DOCUMENT TRACKING SYSTEM

Supporting Documentation

(System Overview, Process Flow, and Data Structure)

1. System Overview

The Document Tracking System is a centralized web-based application designed to record, monitor, and manage the lifecycle of documents across organizational sections. The system provides structured visibility into document ownership, movement, and status transitions.

The platform supports end-to-end document handling, including:

- Document registration and metadata capture
- Document routing and forwarding
- Receiving and acknowledgment
- Status updates (ACTIVE / RELEASED / ARCHIVED)
- Event logging and audit trail
- Attachment management
- Role-based access control

The system is designed around two core operational principles:

Traceability – Every significant document interaction is stored as an immutable system event.

Accountability – Each document movement is recorded as a route between sections with timestamps and responsible users.

2. System Objectives

The system aims to:

- Improve workflow transparency
- Reduce manual tracking errors
- Provide real-time document visibility
- Strengthen document accountability
- Maintain a verifiable audit trail
- Support structured inter-section coordination

3. Core Functional Modules

A. Document Registration

Captures document metadata including requester, document date, subject, content type, communication type, and origin section.

B. Document Routing / Forwarding

Records document transfers between sections using an open-route model.

C. Receiving / Acknowledgment

Closes pending routes and establishes holder accountability.

D. Status Management

Controls document lifecycle states:

- ACTIVE
- RELEASED
- ARCHIVED

E. Event Logging (Audit Trail)

Stores immutable document actions:

- Created
- Sent
- Received
- Forwarded
- Updated
- Released

- Archived
- Cancelled

F. Attachment Management

Supports validated uploads with file-type and size constraints.

4. Process Flow

A. Document Creation + Initial Routing

1. User submits document metadata.
2. System generates a unique tracking number.
3. System inserts document record.
4. System creates an open route.
5. System registers document participants.
6. System logs creation and routing events.
7. Optional attachment validation and storage.

B. Receiving

1. Receiving section views pending routes.
2. Document is acknowledged.
3. Route is closed.
4. Event log updated.

C. Forwarding

1. Current holder forwards document.
2. System creates new open route.
3. Event log updated.

D. Release / Archive

1. Document status updated.
2. System logs lifecycle event.

5. ERD (Entity Relationship Design)

Entity	Purpose	Key fields
Documents	Core document record containing metadata and lifecycle status	tracking_no, requester, document_date, subject, content_type, comm_type, current_status, origin_section_id, current_holder_section_id, created_by_user_id, created_at, updated_at
Routes	Tracks document movement between sections	document_id, from_section_id, to_section_id, sent_by_user_id, sent_at, received_by_user_id, received_at, remarks, cancelled_by_user_id, cancelled_at, is_open
Document Events	Immutable audit trail capturing system actions	document_id, event_type, actor_user_id, actor_section_id, from_section_id, to_section_id, payload_json, created_at
Divisions	Organizational grouping of sections	id, name, is_active, created_at
Sections	Organizational units participating in document workflows	id, name, division_id, is_active, created_at
Users	System actors governed by role-based permissions	id, full_name, email, password_hash, role, section_id, is_active, created_at

Document Participants	Controls document visibility across sections	document_id, section_id, first_seen_at, added_via, added_by_user_id
Attachments	Stores files linked to documents	document_id, original_name, stored_name, stored_path, mime, size_bytes, note, is_append, is_deleted, uploaded_by_user_id, uploaded_by_section_id, uploaded_at

6. Relationship Summary

- Documents → Routes (One-to-Many)
- Documents → Document Events (One-to-Many)
- Documents → Attachments (One-to-Many)
- Documents ↔ Sections (Many-to-Many via Participants)
- Users → Document Events / Routes

7. Role Matrix

Administrator	Division / Section User	Division / Section Viewer
• Full system access • User management • Section management • Configuration control	• Receive / Forward • Status updates • Tracking visibility	• Tracking visibility

8. Implementation Characteristics

- Centralized database architecture
- Role-based access control
- Immutable audit trail
- Open-route movement model
- Attachment validation mechanisms

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